

Weekly Report  
Peter-Emil Georgiev  
Prof. Rao  
Senior Project

This week, I am getting started on choosing the exact algorithms that I will be implementing for my senior project. I have stopped at three:

- implementing a catmull-rom spline for color adjustment ,
- gaussian blur
- seam carving — this will be more complicated to make.

More about the algorithms, which links with information about them is at the bottom of the report.

I also wrote a first draft of the Introduction section of the thesis, which is in the GitHub repo.

Additionally, I will be implementing version control and for that I will also need to implement a graph data structure that will store all changes made. As such, I would also like to implement a merging algorithm, that would allow the merging of multiple unrelated image versions.

For next week, I am going to be working on streamlining my ideas for the project further and starting to work on the app itself — I would like to have a app which can import images , as I want to be able to test the algorithms as I implement them.

Questions for next meeting:

- Are these algorithms enough? This seems like it's too little, while I might be underestimating how difficult they will be to implement.
- How should I go about doing the GUI part? Since it's pretty essential to the essence of my project should I care about it or can the code for that part be neglected?
- When looking to add new features, should I be looking to add a few complex algorithms or should I look at adding many features, so the app looks more complete, but have them be simpler, although mathematically complex?
- While this is a grand project, I notice that the projects that you have provided have more 'altruistic' motivations. The idea behind mine came because I wanted to mix a more artistic hobby with the computer science senior project. Is that enough? Should I look to have a more specific purpose to my project?

Github repo:  
<https://github.com/emolindar/SeniorProjectPemo>

### **Information about the Algorithms:**

Catmull-rom spline:

While general color adjustments, i.e contrast, highlights and shadow adjustments are generally pretty simple to create, as they are mostly just linear transformations of the matrix, which is the image, which is why I will be making it a bit more complicated through adding an adjustment curve. This will be done using a Catmull-rom spline, which, from what I gather is generally what is used. Additionally I can use that for resizing images as well, which would be a 2D extension of the algorithm.

What is a Catmull-rom spline? It is a curve interpolation algorithm, which creates a continuous curve from a few 'checkpoints', which in my case would be the adjustment nodes ( i.e. exposure, highlights and shadows). In resizing, this follows the same principle, but the 'nodes' would be the old pixels, and the smooth curve would be the new pixels that need to be added in order to make the resizing look smooth.

<https://www.cs.cmu.edu/~fp/courses/graphics/asst5/catmullRom.pdf>

## 2. Gaussian blur:

I will also be implementing a Gaussian blur brush. The math for it is listed here:  
<https://www.w3.org/Talks/2012/0125-HTML-Tehran/Gaussian.xhtml>

I will also be utilizing a similar algorithm to use for light normalization across the image.

## 3. Seam carving algorithm for image resizing

Identifying the least important pixels ( seams) and removing them for the purpose of image resizing. This will be more complicated to make efficiently.

<https://www.geeksforgeeks.org/python/resize-an-image-using-seam-carving-algorithm-and-streamlit/>